

CO2-AT3-Capstone Review-1

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QUESTIONS

Q1
Project Proposal
20 marks

Q2
Existing System Analysis
20 marks

Q3
Literature Survey
20 marks

Q4
System Design
20 marks

Q5
Project Planning & Presentation
20 marks

5/5 answered

Q1 Project Proposal

20 marks

Present your capstone project proposal by explaining:• Project Title• Problem Statement• Objectives• Scope of the Project• Expected OutcomesDeliverables:1. Project Proposal Document2. PPT (5–10 slides)3. Initial Architecture Diagram

Your Answer

Project Title:

AI-Powered Food Waste Reduction and Donation Management System

Problem Statement:

Large quantities of edible food are wasted due to poor forecasting of surplus, lack of real-time coordination between donors (restaurants, households, supermarkets) and recipients (NGOs, shelters, community kitchens), and delays in donation scheduling. Additionally, ensuring food safety, pickup timing, and matching suitable food items to recipient needs is often manual and inefficient. This results in both financial loss and preventable environmental impact.

Objectives:

1. To Reduce food waste by accurately predicting surplus availability using AI.
2. Connect food donors with verified NGOs through a real-time donation management workflow.
3. Improve matching of food type, quantity, and timing to recipient requirements.
4. Provide an organized platform for donation requests, pickup scheduling, and tracking.

Scope of the Project:

1. Donor module: Register, list available surplus food items with quantity, category, prepared time, and pickup window.
2. NGO module: Create needs or requests, set preferences (food type, servings, dietary restrictions), and view eligible donations.
3. AI prediction: Forecast donor surplus based on historical patterns (time, location, event-based trends).
4. Matching & recommendation: Rank donations for recipients based on availability time, and suitability.
5. Safety & notifications: Expiry, pickup reminders, and status tracking (requested → approved → picked up → completed).
6. Admin module: Manage user verification, oversee safety guidelines, and monitor system usage.

Expected Outcomes:

1. Decrease in edible food waste by enabling faster and smarter donation matching.
2. Increased utilization of surplus food for community support.
3. Improved efficiency in coordination between donors and recipients.
4. Better transparency and traceability for each donation event.
5. A scalable system that can be deployed for local communities, campuses, and partner organizations.

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Q2 Existing System Analysis

20 marks

Analyze the existing system related to your project. Explain the limitations, challenges, and justify the need for your proposed solution. Include at least three limitations of the existing system.

Your Answer

The existing system for AI-powered food waste reduction and donation management typically relies on manual coordination, basic spreadsheet tracking, and separate communication channels such as phone calls, WhatsApp groups, emails, and donation drop-point logs. Donors (restaurants, hostels, catering services, households) often contact NGOs individually to announce surplus food availability, while receivers (NGOs, shelters, community kitchens) manually confirm pickup/acceptance based on availability. The overall process is not integrated, so coordination depends heavily on human effort and frequent follow-ups.

Limitations of the existing system:

1. High food waste due to delayed communication: This increases the chance that food becomes unusable, leading to avoidable waste.
2. Lack of accurate matching between donors and receivers: Existing systems usually do not consider key attributes such as quantity, type of food, quality/safety score, expiry time, pickup location, and delivery urgency.
3. No centralized tracking: Many current approaches store information in separate spreadsheets or messages. This creates difficulty in tracking donations.
4. Limited quality and safety checks: Existing donation processes often rely only on general statements ("fresh food available") without structured checks. Without standardized verification, food safety risks increase and receivers may discard donations that do not meet their requirements.

Challenges in the existing system:

1. Coordination across multiple parties is difficult because communication is scattered across channels.
2. Maintaining consistent data (food type, quantity, pickup time) is difficult without standardized forms and automated validation.
3. Ensuring food safety compliance is challenging due to limited structured scoring and verification.

Justification for the proposed solution (why an AI-powered system is needed):

The system is required to reduce waste and improve donation success rate through intelligent matching and automation.

The proposed system will:

1. Enable real-time listing of surplus food with structured details (quantity, food type, availability time, location).
2. Use AI-based recommendations to match eligible donors with suitable NGOs based on proximity, urgency, and food category.
3. Reduce human workload by automating request handling, notifications, and status updates.
4. Improve food safety by providing standardized quality/safety assessment and validation before acceptance.
5. Provide better tracking and reporting through centralized records, supporting auditability and performance analysis.

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Q3 Literature Survey

20 marks

Review at least five research papers related to your project. Prepare a comparison table showing the title, methodology, advantages, limitations, and research gap.

Your Answer

- 1."AI-driven Demand Forecasting for Food Retailers" - Focuses on predicting customer demand using ML to reduce overproduction.
Comparison: Similar to your system's waste reduction goal, but it mainly optimizes inventory; it does not fully manage donor-to-donation workflows like scheduling, matching, or real-time approvals.
- 2."Computer Vision for Food Quality and Safety Classification" - Uses cameras/sensors to detect spoilage and classify food safety levels.
Comparison: Strong match for donation eligibility. Your system can adopt this approach to decide which foods are safe to donate, but you additionally need donation management (allocation, logistics, pickup coordination).
- 3."IoT and Smart Waste Monitoring for Restaurants" - Tracks waste generation using sensors and provides analytics.
Comparison: Useful for measuring waste sources and helping continuous improvement. Your system is broader because it not only monitors waste but also triggers AI-assisted actions for donation and coordination.
- 4."Recommender Systems for Matching Food Donations to Recipients" - Uses matching algorithms to connect organizations based on needs and availability.
Comparison: Closely aligned with the donation management part. However, your system can add prediction + quality checking to ensure only suitable food is donated and to improve timeliness.
- 5."Optimization of Donation Logistics Using AI" - Uses routing/scheduling techniques to reduce pickup time and cost.
Comparison: Complements your donation workflow. While it improves logistics, your system also needs intelligent forecasting and eligibility screening to reduce waste effectively from the start.

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Q4 System Design

20 marks

Present the design of your proposed system by including any three of the following:• System Architecture , Datasets

Your Answer

1. Food Waste Records: This dataset contains historical information about wasted food such as food category, quantity, date/time, and reasons for waste (overstock, spoilage, overproduction). It helps the AI learn patterns of what leads to food wastage and when it typically happens.
2. Expiry/Shelf-life and Quality Deterioration Dataset: This dataset provides shelf-life details of different food items along with how quality degrades over time. It enables the system to estimate the safe donation window and prioritize items that are closer to becoming unusable.
3. Donation and Inventory History Dataset: This dataset includes past donation logs including donor/recipient details, food type, quantity, pickup/delivery time, and outcome (successful donation, delayed pickup, or expired). It improves the matching recommendations and increases the likelihood of successful donations.
4. Storage and Handling Dataset: This dataset records storage conditions such as refrigeration/temperature range, time since preparation, packaging type, and handling duration. It supports food-safety risk prediction so donations are made before spoilage.
5. Recipient Demand Dataset: This dataset captures recipient/NGO requirements such as preferred food categories, required quantities, available pickup times, and location. It helps the system match surplus food with the right recipients efficiently and quickly.

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20 marks

Present your project implementation plan including module breakdown, Gantt chart, technology stack, expected timeline, and future work. Demonstrate your initial prototype (if available) and answer the review committee's questions.

UPLOAD PDF FILES * (MAX 2 FILES · 10 MB EACH)

**File 1**

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**File 2 (optional)**

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